

On-site insights: good practices for CRE bullet loan lenders

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Bullet loans, also known in the United States as “interest-only loans”, are a common repayment format in the corporate portfolios of both banks and non-bank lenders, particularly in the context of commercial real estate (CRE) financing.^[1] Because the repayment schedule for bullet loans typically consists of interest-only payments with no principal amortisation during the life of the loan, lenders are exposed to heightened refinancing risk at maturity if borrowers fail to effectively liquidate the asset put up as collateral or secure alternative repayment sources for the capital. In CRE financing, bullet loans are the norm for development financing. They also play a major role in the financing of existing assets (income-producing real estate), property management companies and real estate investment vehicles such as real estate investment trusts.

In recent years, CRE bullet loans have come under increased scrutiny owing to the rapid rise in interest rates and the structural changes that have affected certain asset classes, such as office buildings. Against this backdrop, over the past three years, following a risk-based approach, the CRE on-site inspection campaign has paid particular attention to bullet loans and loans with high balloon payments, where the payment due at maturity accounts for more than 80% of the principal amount borrowed. This article aims to highlight good practices in credit risk management, identified among issuers of these two types of loan in the current environment.^[2] It complements the previous newsletter articles, [“Commercial real estate: connecting the dots”](#) and [“Commercial real estate valuations: insights from on-site inspections”](#).

Mitigating refinancing risk begins with clear definition and measurement

Refinancing risk is the inherent corollary of bullet loans. This risk is particularly prevalent given the typically short tenors involved – around three to five years for income-producing real estate and 18 to 36 months for development projects. Refinancing risk is also procyclical: it tends to remain low during strong market conditions but can increase sharply and abruptly during prolonged downturns, especially when rising interest rates coincide with falling values of assets held as collateral. In such conditions, both lenders and borrowers may find it difficult to sell collateral assets owing to limited market liquidity. Furthermore, during downturns, lenders are often reluctant to refinance loans they did not originate themselves. As part of our CRE on-site inspection campaign, we have identified the following good practices for measuring and mitigating refinancing risk.

1. Banks adopt a clear and explicit definition of refinancing risk for their CRE portfolios and are able to adequately measure it. Refinancing risk is, first and foremost, anchored in the bank's own lending and credit risk monitoring policies. Such policies also take into account the risk appetite of other lenders in the market, which is particularly important for institutions whose risk appetite exceeds that of their competitors. For such lenders, the fact that a loan's key risk indicators meet the bank's internal criteria does not necessarily mean the loan is not exposed to significant refinancing risk. Lenders can acquire deeper insight into prevailing market norms, particularly when taking part in syndication deals.
2. Lending policies mitigate refinancing risk from inception. The most advanced banks conduct a comprehensive repayment capacity assessment, taking the useful economic life of the property into account. They evaluate how long it would take to repay a bullet loan based on available and sustainable cash flow. While debt amortisation scenarios can vary by property type, it is considered good practice for CRE property loans to be repayable over a period of 15-30 years from current and forward-looking cash flows. Special attention is paid to estimating capital expenditure, as this is essential for meeting tenants' anticipated space requirements and upgrading the energy efficiency of financed properties. Leading banks also incorporate other important metrics into their repayment capacity assessments, such as vacancy rates and the weighted average unexpired lease term, alongside sensitivity analyses that account for adverse macroeconomic assumptions. In the recent past, some financial institutions have assumed that interest rates would remain low for several more years. With respect to real estate investment trusts, repayment capacity assessments take account of liquidity pressures arising from the risk of accelerated share redemptions, as well as the objective of making annual dividend payments serviced by a diversified pool of property investments. Lastly, the tenor of loans is calibrated to provide lenders with a reasonable repayment perspective at maturity. This is important to enable proper loan pricing and accurate calculation of expected credit losses.
3. Loans are subject to carefully considered performance covenants. This helps identify potential difficulties in a timely manner, so that effective remedial actions can be taken. Given the time lag in updating valuations, leading banks do not depend exclusively on covenants tied to

metrics like loan-to-value (LTV) ratios. They additionally integrate metrics based on cash flows, as these are inherently more responsive to rising interest rates or deteriorating property performance.

4. Refinancing risk is assessed on an ongoing basis and encompasses all exposures, both as part of periodic reviews of individual loans and in portfolio analyses. The most advanced banks have implemented heatmaps to monitor the refinancing risk of their portfolios, using a traffic light system based on tangible risk metrics such as the LTV ratio, the debt service coverage ratio, the weighted average unexpired lease term, vacancy rates or other relevant metrics. All bullet loans, regardless of their residual maturity, fall within the scope of these heatmaps, although a more in-depth analysis is naturally performed for loans with less than 18 months of remaining maturity. The refinancing risk is analysed both with and without taking potential support from the sponsor into account. The evaluation also includes the necessary down payments to right-size the loan at maturity, ensuring the loan amount meets the bank's lending criteria if asset performance or financing conditions worsen. Refinancing risk is explicitly considered as a criterion for identifying exposures to be flagged for the watchlist or included in unlikelihood-to-pay assessments.
5. Refinancing risk is appropriately reflected by the credit risk models used for monitoring purposes and for calculating provisions. These models consider the repayment schedule, differentiating between bullet and amortising loans. Refinancing risk is identified as a risk driver and assessed using a variety of forward-looking metrics. These include not only the LTV ratio, but also cash flow indicators in the case of income-producing real estate, as well as cost trends and future sales projections for development projects. Furthermore, in line with previous good practice, models account for refinancing risk throughout the entire term of the loan, rather than focusing solely on the year before maturity.
6. Lenders carefully scrutinise key risk driver values, including collateral value and forecasted net operating incomes, to avoid any optimistic bias. Models consider the potential impact of increases in interest rates and credit margins on repayment capacity at maturity, regardless of whether loans are fully hedged for their entire duration. Finally, banks ensure that the data used for the calibration of risk parameters – often based on an extended period of low interest

rates – are representative of the current interest rate regime and of the structural changes affecting certain asset classes, such as office buildings. If not, they consider applying overlays for the calculation of expected credit losses.

Table 1
A case study on refinancing risk

(figures in EUR millions)

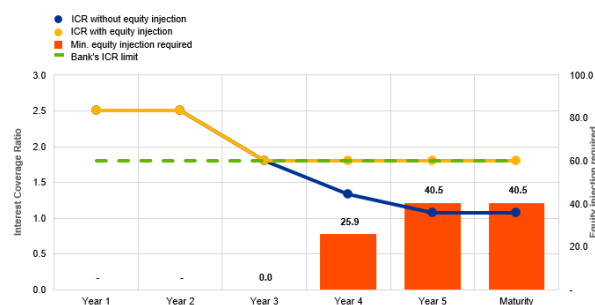
		Inception	Year 1	Year 2	Year 3	Year 4	Year 5	Maturity
Loan repayment characteristics	Loan repayment		4	4	5	6	7	107
	<i>Loan principal repayment</i>		0	0	0	0	0	100
	<i>Interest expenses</i>		4	4	5	6	7	7
Key risk metrics	Collateral value	182	182	182	150	128	115	111
	Interest rate paid by the borrower	4.0%	4.0%	4.0%	5.0%	6.0%	7.0%	7.0%
	Net Operating Income	10	10	10	9	8	7.5	7.5

Source: ECB Banking Supervision.

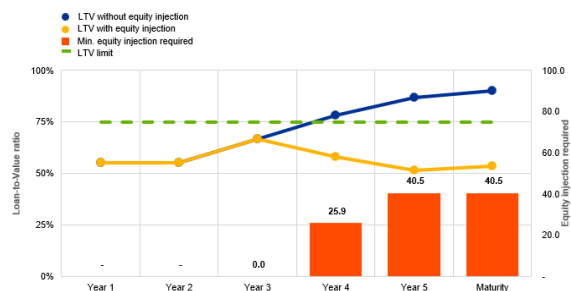
Chart 1

Evolution of key credit risk metrics with and without equity injection

Evolution of interest coverage ratio (ICR) with and without equity injection



Evolution of loan-to-value (LTV) ratio with and without equity injection



Source: ECB Banking Supervision.

Background and explanation of the example

In this fictitious example, an SPV secured a six-year loan of €100 million to finance an office complex. The sponsor of the SPV made an initial equity investment of €100 million, resulting in an LTV ratio of 55% and an interest coverage ratio of 2.5x at loan origination.

By year three, the SPV faces increased interest expenses owing to rising interest rates and a decline in net operating income caused by higher vacancy rates. Simultaneously, the estimated value of the office complex used as collateral decreases from €182 million to €150 million. Under these conditions, the key hurdles set by the bank in its loan origination policy for CRE – namely, a maximum LTV ratio of 75% and a minimum interest coverage ratio of 1.80x – would not be met by year four if the borrower were to refinance the loan with the same bank without an equity injection. For the sake of simplification, this example assumes that lending criteria allow for sufficient cash expenditure given the useful economic life of the property.

Refinancing risk continues to increase in subsequent years owing to further declines in collateral values and net operating income, as well as additional rises in interest expenses. At loan maturity, in the absence of any equity injection during the loan term, the LTV ratio would reach 90% and the interest coverage ratio would drop to 1.07x.

To meet the lender's refinancing standards, the sponsor would need to inject €40 million in equity, representing 40% of the initial loan amount. From the bank's perspective, this operation would "right-size" the loan, reducing it from €100 million to €60 million, thereby aligning it with the new market environment and risk parameters.

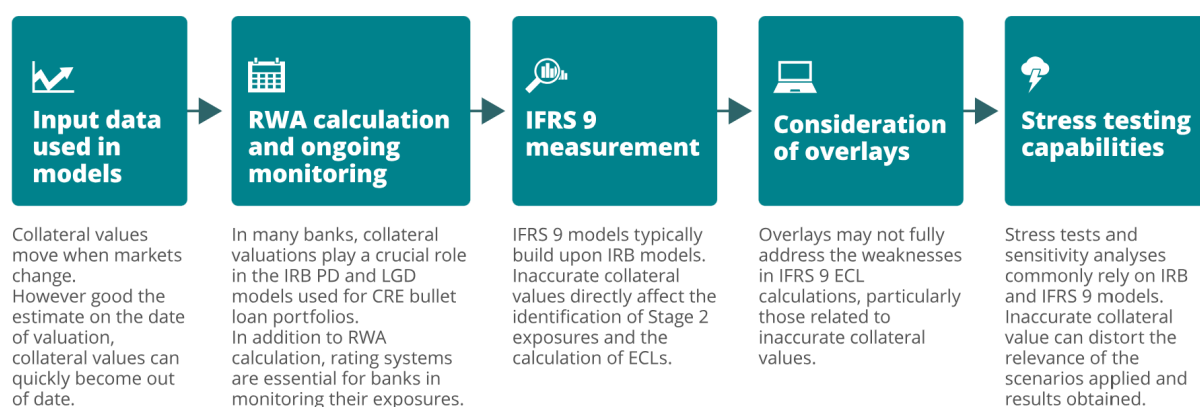
Source: ECB Banking Supervision.

Accurate and robust valuations of collateral and underlying assets are crucial for detecting risk early

For loans secured by real estate collateral, the value of the underlying asset plays a fundamental role in assessing the quality of these loans, especially when it serves as the primary source of repayment. As shown in the graph below, for this type of loan, collateral value is typically a key determinant not only in the loss-given-default models, but also in the probability of default (PD) models banks use to classify exposures into rating grades, given the observed correlation between PD and LTV ratios. Collateral values therefore have a direct impact on the calculation of risk-weighted assets and expected credit losses. They are also pivotal for stress testing and sensitivity analyses of CRE portfolios, which are primarily based on internal rating models. Hence, outdated or flawed collateral valuations can cause ripple effects throughout the lenders' entire risk management and accounting frameworks, impeding the timely identification and mitigation of emerging risks.

Chart 2

Inaccurate collateral values have a ripple effect on credit risk measurement



Source: ECB Banking Supervision.

The value of underlying property assets is also pivotal for loans granted to real estate investment entities, such as property companies or real estate investment trusts, even when the financing is not formally secured by the assets in question. In the absence of additional guarantees, the ability of real estate investment entities to repay the loan largely depends on the performance of these assets and subsequent changes in their market value. The following section shares good practices identified in the current environment for valuing assets received as collateral.

1. Leading lenders have accelerated the revaluation cycle in the event of abrupt negative changes in macroeconomic conditions, such as an increase in interest rates and credit margins. The revaluations have primarily targeted assets that have experienced significant changes in occupiers' requirements, as has been seen with prime offices in recent years. The most advanced lenders conducted new valuations even amid reduced transaction volumes

during the market downturn. This reduction was interpreted as both a potential decline in market value and a factor contributing to increased uncertainty in most valuations. Such uncertainty was assessed by means of sensitivity analyses.

2. In stable market conditions, the most advanced lenders perform a full revaluation at least every three years for loans exceeding €3 million, or 5% of the own funds of an institution. The three-year limit laid down in Article 208(3) of the Capital Requirements Regulation is regarded as a backstop. Compared with a simple review of an existing valuation, a full revaluation of a property involves conducting a comprehensive assessment of the asset's value, including an on-site visit, a thorough analysis of current market data, and consideration of all relevant factors affecting the property's market value. From that perspective, most banks regard a full revaluation as providing a more robust and up-to-date estimate of a property's market value.
3. With respect to development projects, leading lenders request independent valuers to provide sensitivity analyses both at loan inception and during subsequent reviews as the project progresses. These analyses model the effect of changes in both the expected remaining costs to complete and the current market value of the property as if complete. Even though the principal in a bullet loan is based on the value as if complete, the most advanced lenders regularly monitor the current value of the asset under construction or development throughout the project to remain fully informed about the viability of the project and the risks associated with the loan.
4. It is also regarded as good practice for the principle of rotation to be applied not only to individuals, but also to valuation firms whenever possible. Rotating appraisers within the same team may not be sufficient to adequately address the threat of confirmation bias, as individuals from a single company will naturally collaborate and discuss their perspectives on each valuation processed. The avoidance of confirmation bias is also the reason many regulators require entities to regularly rotate their auditor companies. When valuations are carried out in-house or through an entity controlled by the bank, the most advanced lenders benchmark a selection of these valuations against those produced by an external valuer with recent experience in the relevant market.

5. Lenders formally appoint valuers, where possible. They also ensure they are compensated directly and fairly, with no connection between the amount of compensation and the valuation outcome. The most advanced lenders make sure that the appraiser's invoice is addressed to them. They avoid situations where valuation invoices are sent to the borrower, even if the valuer has been appointed by them. Indeed, a dissatisfied borrower may be inclined to significantly delay payment, which could undermine the independence and objectivity of the valuation process.
6. Records of discussions and correspondence involving the appraiser, the lender and the borrower are maintained to help safeguard and demonstrate the valuer's independence from those responsible for commissioning the valuation. This is especially important if any changes are made to a draft valuation following such interactions. Although the independent valuer may have received additional information that justifies a change in their initial assessment, without a clear and documented record an auditor cannot verify that the change was not influenced by pressure from the lender or the borrower.
7. Banks have a suitably qualified team to review the valuations provided by the valuer, ensuring they comply with the applicable professional standards issued by professional bodies such as the International Valuation Standards Council, the Royal Institution of Chartered Surveyors and the European Group of Valuers' Associations. This team operates independently from the business line and critically assesses any aspect of the valuation that appears unclear or inappropriate. They may report cases of clear non-compliance to the relevant professional bodies.
8. Banks ensure that the real estate investment entities they finance, whether public or private, adhere to best practice codes for valuations established by trade associations such as the European Public Real Estate Association or the European Association for Investors in Non-Listed Real Estate Vehicles. These best practice guidelines are, for the most part, similar to those established for the valuation of property assets used as collateral in secured lending. They address key aspects such as the conditions for the selection and appointment of valuers, the principle of valuer rotation, and the valuation process itself. For example, the European Public Real Estate Association's Best Practices Recommendations Guidelines recommend that

companies include in their financial reports either a summary of the valuation report approved by the valuer or a reconciliation table linking the valuer's reported amounts to those included in the financial statements. If there is a clear failure to implement guidelines recognised as best practice in the relevant sector, lenders may ask the borrower to explain the reasons for this and consider the risks associated with non-compliance.

9. In addition, lenders make full use of available information to challenge the valuation practices of real estate investment entities, including funds. Specifically, for assets they finance directly through a special purpose vehicle, they compare their own valuations for collateral assessment purposes with those reflected in the consolidated statements of the special purpose vehicle's parent company. They also check that valuation changes reported in the financial statements of investment entities are consistent with underlying market conditions. Furthermore, lenders verify that the sales prices achieved align with the market values reported by the entity prior to the respective sales transactions. More broadly, banks with a better understanding of the valuation practices of real estate investment entities are better equipped to tailor their risk management approaches to the entities they finance.

Deeper insight into sponsors is critical for banks assessing the resilience of CRE bullet loan portfolios in challenging times

Sponsors are the cornerstone of CRE bullet loan portfolios, even when the loans are non-recourse. Ultimately, in the event of a market downturn or challenges specific to the investment, only the sponsors are able to inject additional equity or provide guarantees to facilitate the refinancing of the loan. However, in the absence of formal guarantees, even a top-tier sponsor may still choose to “hand back the keys” to the lender despite the potential reputational risk. Given that changes in macroeconomic conditions and structural shifts in certain sectors have made it increasingly difficult to refinance certain projects or assets on a standalone basis, a thorough understanding of sponsors' capacity and willingness to provide additional support is crucial for lenders. The following section shares good practices identified during the CRE on-site inspection campaign regarding the consideration of sponsors, from loan origination all the way through to the unlikeliness-to-pay assessment.

1. Granting criteria include explicit requirements regarding sponsors. In line with the EBA Guidelines on loan origination and monitoring, “when the borrower is a special purpose vehicle sponsored by another entity, institutions should assess the sponsoring entity's experience in relation to the type, size and geographical location of the CRE”. Additionally, it is considered

good practice for lenders to consider the sponsor's track record in supporting financings that have encountered difficulties in the past.

2. Over the life of the loan, leading lenders check that sponsors have both the capacity and the willingness to inject additional equity if needed, as part of both individual exposure reviews and portfolio analyses. To this end, lenders conduct cash flow analyses that consider all contingent funding obligations undertaken by the sponsors, as well as their ability to inject capital, including in adverse market conditions. Even if sponsors are deemed capable of supporting distressed investments and qualifying for refinancing, the most advanced lenders assess whether it would be contractually required or economically rational for a sponsor to commit additional funds ("good money after bad") under such circumstances. Since sponsors may have reduced their "skin in the game," as they tend to recoup their initial investment during the early years of financing through dividends or management fees, some lenders have proactively engaged with sponsors to gain greater insight into the outcomes of their strategic portfolio reviews. When the sponsor's potential support appears inadequate, lenders regard this as an early warning indicator at least.
3. Banks establish clear rules on how they take sponsor support into account in their unlikelihood-to-pay assessment. These rules have a strong forward-looking component, which includes the consideration of potential sponsor support. In the context of bullet loans, a lender cannot assume that a sponsor will continue to support a problematic investment solely based on the fact that the sponsor has previously invested a significant amount of equity and continues to cover interest payments or ongoing expenses related to the asset. In leading banks, the consideration of sponsor support in the unlikelihood-to-pay assessment is subject to a thorough and properly documented analysis of the sponsor's capacity and willingness to inject additional equity, as outlined in the previous paragraph. Loans that would otherwise be classified as non-performing in the absence of sponsor support are clearly identified and carefully tracked at the portfolio level.
4. Leading banks do not limit their review to the direct ownership structure of the entity they finance but extend the scope of their analysis to the larger group or network to which the direct owner is connected, particularly in the case of real estate management groups. When the

sponsor is an investment fund, it is considered good practice for lenders to have an aggregated view of the financing granted involving the asset management group. This aggregated view helps to highlight any potential dependency risk on these asset management groups and possible contagion effects. It also enables leading banks to gain a better understanding of the practices these groups adopt in dealing with problematic assets. It is also regarded as good practice for banks to broaden their sponsor review to cover junior lenders, as they can play a key role in supporting borrowers undergoing restructuring.

Holistic information is essential for bank boards managing CRE bullet loan portfolios

Governing bodies are responsible for setting the strategic direction of the bank. They must be able to identify vulnerabilities at an early stage in order to act proactively, ensuring they stay ahead of challenges rather than reacting to them after the risk materialises. This is especially pivotal for banks with a significant CRE bullet loan portfolio, as they must navigate uncertain market conditions.

Transparency and informed decision-making are essential in these circumstances.

The following good practices primarily involve banks' management bodies and affiliated risk committees.

1. Board members are made aware of the limitations of key risk metrics in the event of rapidly changing market conditions. This is especially relevant for bullet loans secured by CRE properties, where collateral valuations play a critical role in determining LTV ratios and can also significantly impact the calculation of expected credit losses. During the interest rate hikes of 2022-23, the most advanced banks provided their board members with indicators to assess the reliability and timeliness of the valuations used by the bank in its metric calculations. Particularly for adjusted collateral valuations, board members were also equipped with information allowing them to distinguish the proportion derived from fully-fledged revaluations versus those based on a simple review of an existing valuation.
2. Board members are provided with more detailed information regarding refinancing risk. This information goes beyond maturity walls, which depict the percentage of debt maturing in the near term, and also includes data on changes in the share of loans refinanced by the bank at maturity. An increase in this proportion could indicate that refinancing risks are materialising. Taking a more forward-looking approach, leading banks identify the proportion of CRE bullet loans that are unlikely to meet refinancing criteria at maturity without sponsor support. As a

good practice, this analysis is grounded in the results of sensitivity analyses that stress key parameters, such as the debtor's cash flow and collateral value. Finally, board members have access to both qualitative and quantitative information regarding the capacity, willingness or contractual obligation of key sponsors to support loans reaching maturity.

3. Board members discuss the level of provisions related to CRE bullet loan portfolios. The overall provision amount cannot solely result from the application of models, especially when dealing with novel risks such a high-interest rate environment or structural shifts that accelerate the obsolescence of properties like offices. Like any tool, these models can have limitations stemming from their design and the data used for their development, and such limitations are clearly communicated to the board members, especially in a context of rapidly changing market conditions. Validation teams help ensure that overlays used to adjust the expected credit losses are carefully calibrated to reflect current economic conditions or specific risks that the models might not fully capture.
4. Board members ensure that exposures that have been subject to a significant increase in credit risk are appropriately reported in IFRS 9 Stage 2. Where applicable, they ensure that a top-down mechanism, i.e. at the level of a sub-portfolio, has been implemented to address the weaknesses in identifying these exposures at an individual level.
5. Board members explicitly address and discuss the risks associated with “amend and extend” approaches which have facilitated the development of broader plans to address the most vulnerable components of CRE bullet loan portfolios from a medium-term perspective. This strategy may include selling problematic loans, increasing provisioning ratios, or strengthening portfolio-level limits governing CRE bullet loans. It was an understandable choice to rely on “amend and extend” approaches for problem loans during the initial phase of shifting monetary conditions, where lenders frequently focused on extending loan maturities and seeking cash injections from sponsors whenever feasible. However, in the absence of a significant recovery in real estate markets, sponsors may become less willing or able to provide additional cash injections during the next wave of refinancing, particularly amid increased demand for capital and ongoing geopolitical uncertainties.

Many of the good practices highlighted in this article are not specific to CRE. They can be effectively applied to any type of financing involving bullet loans, particularly in the context of asset, acquisition or project financing, regardless of the lender's regulatory status or jurisdiction.

The authors wish to express their gratitude to the numerous inspectors, as well as the joint supervisory and horizontal teams involved in the CRE campaign, for their contributions to the article and its review.

1.

This article focuses on CRE financing, specifically lending in the context of income-producing real estate as well as CRE properties that are under development or construction.

2.

It does not establish supervisory expectations for best practices that go beyond current regulations.

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